

Isaac Nason

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Printables: printables.com/@NasonNation_3713912

Professional Summary

Senior Embedded Software Engineer with 11 years of experience bridging high-level software architecture (Python, C#, .NET) with physical hardware. Specialized in rapid mechatronic prototyping, microcontroller programming (ESP32, RP2040), and sensor fusion. Proven ability to architect custom electromechanical systems, managing the entire lifecycle from CAD concept and power distribution to functional deployment.

Core Competencies & Technical Skills

- **Microcontrollers:** ESP32 (C3/C5/S3), RP2040, Raspberry Pi 3A+/4B, Arduino UNO R4, Seeed Studio XIAO.
- **Languages:** Python, MicroPython, CircuitPython, C#, .NET, SQL, JavaScript, HTML/CSS, XAML, Xamarin.
- **Hardware:** I2C, SPI, UART, motor encoders, actuators, solenoids, tactile exciters, voltage step-down converters, FPV telemetry.
- **Tools:** Git, Jira, Azure DevOps, CI/CD, Agile Methodologies, IoT Telemetry, Firebase.
- **CAD & Fab:** SolidWorks (learning), 3D Builder, Blender, Bambu Lab X1 Carbon, rapid iteration manufacturing.

Mechatronics & Embedded Engineering Experience

Independent Mechatronics Engineer | NasonNation Robotics | Cornelius, OR

Aegis Control Sentry Turret (Published Oct 2025)

- Architected an AI-powered physical sentry turret, utilizing Roboflow computer vision and Python control logic for real-time targeting and mechanical tracking.
- Engineered custom power distribution to isolate logic and actuator rails, and developed a Flask web interface for live telemetry monitoring and remote overrides.
- Designed and 3D-printed custom mechanical housings, managing material tolerances and actuator clearances.

Haptic VR Controller System / Sentient Sidearm (Sep 2025 – Dec 2025)

- Engineered a custom haptic feedback system (submitted to the Hackster.io Sensor Fusion Challenge) integrating tactile exciters, solenoids, step-down converters, and audio power amplifiers.
- Wrote embedded control scripts to synchronize physical force-feedback with real-time VR telemetry data.

UAV / FPV Flight Systems (Aug 2026)

- Built and tuned over 12 custom multi-rotor drones, integrating analog FPV systems, 1S-6S LiPo power delivery configurations, and flight telemetry.

Professional Software Systems Experience

Business Analyst Programmer | City of Beaverton | Beaverton, OR | May 2021 – Feb 2026

- Architected enterprise software systems and engineered multi-layer APIs using secure tokenization to handle distributed data access across municipal databases.
- Developed scalable user interfaces (GUIs) for critical public and internal applications using C#, .NET, and SQL Server.
- Streamlined system integrations and automated data workflows, applying engineering standards to ensure zero-downtime deployments.

Senior Software Engineer & Founder | MyAppMobile, LLC | Feb 2015 – May 2021

- Architected and deployed 9 cross-platform edge-device applications (Android/iOS) utilizing .NET, Xamarin, and Google Firebase.
- Maintained custom backend infrastructure, using secure APIs to scale mobile solutions for 800+ concurrent users with 99.9% uptime.

Service Program Specialist (Internal Developer) | Tektronix | Beaverton, OR | Jan 2011 – May 2021

- Engineered high-impact internal tools utilizing the Microsoft Power Platform and SSMS to automate worldwide customer support operations for a global hardware manufacturing leader.
- Designed automated technical workflow processes that saved colleagues an estimated 10 hours a week on manual data entry.

Education & Certifications

- **B.S. Business Management**, Portland State University
- **Certifications:** Criminal Justice Information Services (CJIS) Certified